

COSC 594/650

Constrained optimization

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Convergence analysis of optimization algorithms

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

■ ASYMPTOTIC CONVERGENCE

■ RATE OF CONVERGENCE

■ ITERATION COMPLEXITY ANALYSIS

Method	Convex	Strongly convex
GD	$O(1/\epsilon)$	$O(\kappa \log 1/\epsilon)$
Nesterov	$O(1/\sqrt{\epsilon})$	$O(\sqrt{\kappa} \log 1/\epsilon)$

Linear regression

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	5	4	5	3

$$y_i = \theta^\top x_i$$

Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_n	4

Input: rating features x_i

Output: Chloe's rating y_i

Model: linear predictor $\theta^\top x_i$

Loss: squared loss $(\theta^\top x_i - y_i)^2$

QUESTION: What if we want a more interpretable predictor?

Constrained linear regression

$$\begin{array}{ll}\underset{\theta}{\text{minimize}} & \|X\theta - y\|^2 \\ \text{subject to} & \theta \geq 0 \\ & \mathbf{1}^\top \theta = 1\end{array}$$

- ★ interpretable trust weights θ
- ★ many other constraints

Convex, but constrained!

From unconstrained to constrained optimization

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

$\Downarrow$

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in X \end{array}$$

QUESTION: How should we modify our existing algorithms?

First-order optimality over a convex set

Convex function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

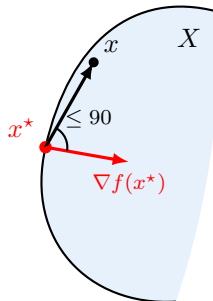
Convex set: $X \subset \mathbb{R}^n$

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in X \end{array}$$

■ LOCAL MINIMUM

$$\langle \nabla f(x^*), x - x^* \rangle \geq 0, \quad x \in X$$

■ GLOBAL MINIMUM



CONVEXITY:

Local minimizers $\equiv$ global minimizers

Two important feasible sets

■ NONNEGATIVE ORTHANT

$$X = \{x \mid x_i \geq 0\}$$

$$x_i^* > 0 \quad \implies \quad \frac{\partial f(x^*)}{\partial x_i} = 0$$

$$x_i^* = 0 \quad \implies \quad \frac{\partial f(x^*)}{\partial x_i} \geq 0$$

■ PROBABILITY SIMPLEX

$$X = \{x \geq 0 \mid \mathbf{1}^\top x = 1\}$$

$$x_i^* > 0 \quad \implies \quad \frac{\partial f(x^*)}{\partial x_i} \leq \frac{\partial f(x^*)}{\partial x_j}, \quad \forall j$$

Feasible-direction framework

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Feasible direction: d_k

Feasible stepsize: $\alpha_k > 0$

■ FEASIBLE DIRECTION

$$\exists \alpha > 0, \quad x + \alpha d \in X$$

■ FEASIBLE STEPSIZE

- ★ $\alpha_k = \alpha$ – constant
- ★ $\alpha_k \in \operatorname{argmin}_{\alpha \geq 0} f(x_k + \alpha d_k)$ – exact minimization
- ★ $\alpha_k \in \operatorname{argmin}_{\alpha \in (0, \bar{\alpha}]} f(x_k + \alpha d_k)$ – limited minimization
- ★ $\alpha_k \rightarrow 0, \sum_k \alpha_k = \infty$ – diminishing

Gradient-related condition

Subsequence: $\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x}$, a non-stationary point

Feasible directions: $\{d_k\}_{k \in \mathcal{K}}$

■ GRADIENT-RELATED DIRECTIONS

- ★ $\{d_k\}_{k \in \mathcal{K}}$ – bounded feasible directions
- ★ $\limsup_{k \rightarrow \infty, k \in \mathcal{K}} \nabla f(x_k)^\top d_k < 0$ – subsequence limit

■ STATIONARY-POINT CONVERGENCE

$$\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x} \implies \bar{x} \text{ is a stationary point}$$

Every limit point x^* is stationary: $\langle \nabla f(x^*), x - x^* \rangle \geq 0, \forall x \in X$

Convexity is not required!

QUESTION: How to choose feasible directions?

Conditional gradient (Frank–Wolfe)

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Feasible direction: d_k

Feasible stepsize: $\alpha_k > 0$

■ CONDITIONAL GRADIENT

$$\bar{x}_k = \operatorname{argmin}_{x \in X} \langle \nabla f(x_k), x - x_k \rangle$$

$$\implies d_k = \bar{x}_k - x_k$$

Direction d_k is well-defined (compact X) & gradient-related

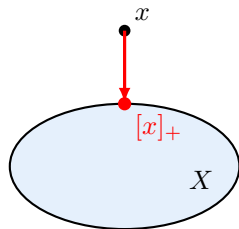
Gradient projection

Iterative update: $x_{k+1} = [x_k + \alpha d_k]_+, k = 0, 1, 2, \dots$

Direction: $d_k = -\nabla f(x_k)$

Stepsize: $\alpha > 0$

■ PROJECTION ONTO A CLOSED CONVEX SET



$$[x]_+ = \operatorname{argmin}_{z \in X} \|z - x\|^2$$

- ★ $\langle x - [x]_+, z - [x]_+ \rangle \leq 0, \forall z \in X$ – necessary and sufficient condition
- ★ $\|[x_1]_+ - [x_2]_+\| \leq \|x_1 - x_2\|, \forall x_1, x_2 \in \mathbb{R}^n$ – non-expansive and continuous

$$x^* \in X \text{ is stationary} \iff x^* = [x^* - \alpha \nabla f(x^*)]_+, \forall \alpha > 0$$

Convergence analysis for projected gradient descent

Strongly convex function: $f(x) = \frac{1}{2}x^\top Qx + b^\top x$, $Q \succeq \lambda_{\min}I \succ 0$

Convex set: $X \subset \mathbb{R}^n$

Stepsize: $\alpha = \frac{1}{L}$

■ ERROR BOUND

$$\begin{aligned}\|x_{k+1} - x^\star\| &= \left\| \left[x_k - \frac{1}{L} \nabla f(x_k) \right]_+ - \left[x^\star - \frac{1}{L} \nabla f(x^\star) \right]_+ \right\| \\ &\leq \left\| (x_k - x^\star) - \frac{1}{L} (\nabla f(x_k) - \nabla f(x^\star)) \right\| \\ &\leq \left(I - \frac{\lambda_{\min}}{\lambda_{\max}} \right) \|x_k - x^\star\|\end{aligned}$$

Iteration complexity: $\mathcal{O} \left(\kappa \log \frac{1}{\epsilon} \right)$

Questions?

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